

more desirable from a query-efficiency standpoint. Another consideration when comparing attacks is that attacks with “many-trial” characteristics are more likely to overstate their performance because they benefit from the classifier’s false positive rate (FPR) for each query. Unless the results are manually verified³, each additional inference raises the risk of sampling an outlier harmful generation or misgrading an unsuccessful attack as a false positive. In addition, recent results show that even with manual verification, it is possible to successfully jailbreak models by simplifying sampling a large number of responses, sometimes without altering the harmful prompt at all, further advantaging many-trial approaches (Hughes et al., 2024). Given these disparities, we argue that these two setups **must be clearly distinguished, and that many-trial attacks should account for the classifier’s FPR when assessing the significance of results**. In addition, wherever possible, but particularly for “many-trial” jailbreaks, **we recommend manually screening generations for false positives**.

The LLM-judge ecosystem is fragmented. Due to the prohibitive cost of human annotators, most works rely on automated classifiers to decide whether an output is harmful. However, the landscape is highly fragmented: some methods train custom classifiers (Huang et al., 2023; Zhu et al., 2024; Dai et al., 2023), others use fine-tuned LLM judges like Harmbench Llama-2 (Mazeika et al., 2024; Bor-eiko et al., 2024; Zou et al., 2024) or StrongREJECT (Souly et al., 2024; Guan et al., 2024), while others employ pre-trained models such as GPT or Llama variants using custom prompt templates (Chao et al., 2024; Hughes et al., 2024; Thompson & Sklar, 2024b). Some approaches incorporate human verification (Zou et al., 2023b; Hughes et al., 2024; Guan et al., 2024), and others fail to specify their evaluation methodology entirely. Although there are good reasons to train a new classifier (e.g., systematic failures in preexisting ones), the disjointed ecosystem of judges, prompt templates, and evaluation strategies complicate meaningful comparisons across studies.

In addition, despite prior work demonstrating that generation length affects attack success rates (Mazeika et al., 2024), there is still no consistent standard in the literature. Some works propose to use a consistent amount of judge tokens, while others use the victim model to generate a fixed number of tokens. While there are arguments to be made for both cases, we think the most realistic threat model would evaluate a full model response up to the end-of-turn/end-of-text token. This is closest to real-world use cases where users chat with models in a (mostly) unconstrained setting. Finally, varying definitions of “successful” jailbreaks are used, with a clear trend of criteria becoming increasingly

³In fact, even human judges may have nonzero FPR and could be affected too.

strict and precise over time, which we see as a positive development. We do not claim to have definitive answers, and we recognize that the community has not yet reached a consensus. Nevertheless, we believe that researchers would benefit from a more standardized judging environment. As judging is relatively compute-inexpensive compared to running attack algorithms or training a model defense, **we recommend that new works report results on multiple popular judges** (e.g., Harmbench Llama-2-13b and StrongREJECT), **and include a detailed description or code for the exact judge setup used** to facilitate easier comparisons and reproductions.

Judges are not verified against new attacks & defenses.

Since attacks and especially defenses can significantly change a model’s output distribution, generated responses may fall far outside the judges’ training data, causing attack- and defense-specific misclassifications.

In Appendix A.5.2, we show that model-specific differences occur in practice and can lead to misleading comparisons between the safety of two models. We also conduct a large-scale study on 5,269,564 generations and compare judgments by Harmbench’s Llama-2 13B classifier and StrongREJECT’s Gemma 2B judge, and reveal attack- and model-specific biases—up to 25% for specific attacks and up to 24% across models (see Figure 3), complicating attack, defense, and model evaluations. Detailed results & analysis are provided in Appendix A.6. Similar biases have been observed before; for instance, Hughes et al. (2024) find that circuit breaker models (Zou et al., 2024) have higher judge FPR compared to baselines, which would lead to unfair comparisons without manual verification.

To avoid misleading results, **we propose that new attacks and defenses should include at least a small-scale human-verification that judges still work for the proposed method**, e.g. by manually checking 100 examples, as is done in (Chao et al., 2023). This is a manageable time investment that ensures attacks and defenses remain comparable and shields from effects such as the higher-than-expected FPR of some judges on specific models.

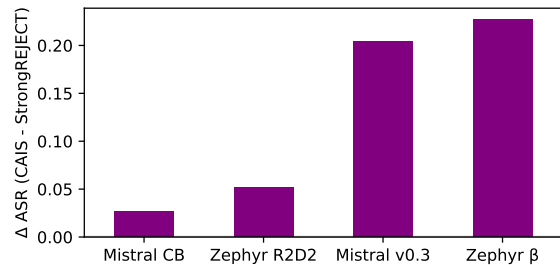


Figure 3: Comparing ASR differences across common judges reveals significant model-specific bias, despite all four models being derivatives of the same Mistral 7B model.

5. Alternative Views

Small datasets have significant practical advantages.

While we advocate for the adoption of larger datasets, smaller ones offer key benefits. Even with small sample sizes, LLM robustness research is computationally expensive due to the size of modern models and the need for strong adversarial attacks to ensure reliable evaluations. Unlike prior adversarial robustness research in fields like computer vision, optimizing attacks in a discrete input space adds further complexity: current approaches require hundreds of thousands of queries per prompt, creating a bottleneck in robustness research. Using larger datasets would only exacerbate these issues and further raise the barrier to entry. Meanwhile, the lower computational demands of small datasets accelerate experimentation and research cycles, particularly in resource-constrained academic settings. This facilitates innovative research, as hypotheses can be evaluated quickly and cost-effectively. For instance, recent work (Andriushchenko et al., 2024) convincingly demonstrated that model-adaptive attacks can reach very high ASR using relatively few (50) samples. In general, even small datasets can provide enough signal to show the usefulness of a new approach at lower cost *if* the effect size is sufficiently large. Even when precise results are required, sophisticated subsampling strategies can be used to enhance the predictive power of small-scale experiments. For instance, Polo et al. (2024) reveal that a carefully selected subset of 100 questions can reliably approximate LLM capability on the full MMLU dataset with over 15k questions. This indicates that larger sample sizes may not be a strict requirement for comprehensive LLM evaluations.

On flawed LLM judges. Compared to other, more constrained domains with well-defined tasks and success criteria (e.g., image classification) the desire to find consistent and objective metrics may simply be misplaced in the context of LLM-robustness evaluations. Language is ambiguous and even our most precise rules will likely never catch all edge cases. The best we can do is iteratively refine our measures of jailbreaking success and continuously adapt automated judge models to capture these definitions more closely. In fact, this is exactly what the field is doing: researchers are constantly developing improved judge models that rely on increasingly precise definitions and better approximate human judgment (Mazeika et al., 2024; Souly et al., 2024; Zhu et al., 2024). Though imperfect, these evaluations are the only practical and scalable approach available.

Perhaps we should explore more fundamental blind spots. As models grow more capable and show better awareness of their condition, they may begin to strategically adapt their outputs to mislead researchers about their true nefarious capabilities. Recent work shows that even current models may deceptively adjust their outputs in response to a fictitious description of an environment that incentivizes

such behavior (Greenblatt et al., 2024). Given these findings, maybe the community should instead move away from black-box input-output evaluations via judges and consider approaches like interpretability (Bereska & Gavves, 2024; Ardit et al., 2024) or representation engineering (Zou et al., 2023a). Understanding the internal mechanisms behind harmful outputs could enable more granular and robust assessments that help reduce the risk of deception.

Finally, empirical evidence suggests that machine learning progress continues despite flawed metrics and noise.

Salaudeen & Hardt (2024) show that architecture improvements developed on a particular dataset are often robust to distribution shifts. Meanwhile, commonly used evaluation metrics for image generation have been criticized for bias and disagreement with human perception (Kynkäänniemi et al., 2022), yet the field has rapidly advanced (Rombach et al., 2021). If imperfect benchmarks & biased datasets were fundamental barriers, such rapid progress would be highly unlikely. Similarly, we continue to see genuinely new and innovative approaches in the context of LLM robustness (Zou et al., 2023b; Yu et al., 2024), contradicting the notion that flawed metrics and noise hinder innovation.

6. Conclusion

In this work, we critically examined the current state of LLM robustness evaluations, highlighting major sources of noise, bias, and inconsistencies across datasets, algorithms, and evaluation pipelines. Our analysis reveals that small and inconsistently subsampled datasets, implementation discrepancies, and fragmented evaluation setups severely hinder the reproducibility and comparability of research in this space.

To address these issues, we recommend the following:

- Larger, high-quality benchmarks should be prioritized to reduce statistical uncertainty.
- Data subsampling should be avoided to reduce the need to reproduce past work and ease cross-paper comparisons.
- The field should work to standardize & carefully document implementation details like quantization, tokenization, and chat templates, as they significantly impact ASR.
- Methods should be compared under fair conditions; attacks should be controlled for effort or success rate, and defenses should consistently report results for safety & overrefusal to ensure meaningful evaluations.
- Finally, to reduce bias in the judging phase, we recommend using multiple judge models, manual verification of judge effectiveness on attacks and defenses, and a shift from greedy generation towards distributional evaluation.

Our vision for LLM-robustness research is a field with correct, consistent, reliable, and reproducible results that are easily comparable across different projects, be it new attack algorithms, defenses, or models. We hope our work serves as a constructive step towards this goal.

Impact Statement

We present work in the field of AI safety and robustness in the context of LLMs. Due to their widespread availability and rapidly increasing capability, their reliability and safety has significant consequences beyond the research community and may be of interest to industry and the general public. Researchers have an obligation to ensure their work meets scientific standards, such as reproducibility, and accurately communicates findings as objectively as possible. Many of our recommendations are centered around making sure that the field has the ability to accurately fulfill these standards without being misled by faulty or noisy evaluations.

Prior work on safety & alignment has also sparked controversy regarding the perceived trade-off between censorship and model safety. We acknowledge that it is possible to misuse alignment technology for nefarious purposes, such as censorship. However, we believe that the problems of alignment/robustness and the specification of what models should be aligned to should be treated as separate. We explicitly avoid making any judgements or recommendations related to the latter, and focus on proposing objective ways to compare true model behavior to a set of guidelines.

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